

- 13** Two monoatomic gases, X and Y, are in thermal equilibrium in a mixture. The molecular mass of gas Y is half the molecular mass of gas X.

The root mean square speed of the molecules of gas Y is v . What is the root mean square speed of the molecules of gas X?

- A** $\frac{v}{2}$
- B** $\frac{v}{\sqrt{2}}$
- C** v
- D** $2v$